Parallelizing the Sequential Agent Loop on Unified-Memory Personal Computers

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Jerry Chen (chinjuic) Ethan Chiu (ethanchi)

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URL

Project page: `https://kogby.github.io/pp-final-project/`
Code: `https://github.com/kogby/pp-final-project`

Summary

We will parallelize the sequential loop of a coding agent (LLM generation → tool execution → LLM generation → ⋯) on two unified-memory machines, an NVIDIA GB10 (DGX Spark) and an Apple M5 Pro laptop, replaying real agent traces on `llama.cpp` with real tools (e.g. `ripgrep`, a compiler, `pytest`, `pandas`). We work at two levels: (1) within one agent, we overlap CPU-side tool execution with GPU-side generation through pipelining and speculation in both directions, though a preliminary estimate suggests this gain may be limited; (2) across agents, we run several independent agent sessions on one machine with a shared, batched decode stream and concurrent tools. We ask how much parallelization gains on a machine where extra work shares memory bandwidth and capacity with the critical path, and how much a scheduler gains when it knows each tool's and agent's workload.

Background

The sequential agent loop. A coding agent (Claude Code, Codex CLI, OpenHands, ...) runs a loop: the LLM generates a response that ends in one or more tool calls, then the harness executes the tools, whose output is appended to the context and the LLM generates again. Each LLM step depends on the previous tool result, and each tool call depends on the previous LLM step, so one request is a long dependency chain. Public traces of real coding-agent sessions (e.g. TraceLab [1]: 357k LLM rounds, 433k tool calls from 43 users) show roughly 8.8 LLM steps and 10.8 tool calls per user request. Tool latency is heavy-tailed that about 4% of calls that run over a minute account for about 85% of tool time.

Four ways to break the chain inside one agent. (Figure 1):

- Pipelining (Conveyor [6]): start executing a tool call as soon as its arguments have been decoded,while the LLM is still generating the rest of its response.
- Speculation (PASTE [5]): predict the next tool call from the pattern of previous calls and execute it during generation. (discard result if no match)

- DAG decomposition (LLMCompiler [7]): execute independent tool calls concurrently.
- Result speculation (value prediction [8]): while a tool runs, fork k generation branches, each conditioned on a likely tool outcome (e.g., build succeeds / fails). When the tool finishes, only keep the branch with correct assumed value.

PASTE and Conveyor hide tool or generation latency behind each other by speculation. All four ways above trade extra work for a potential shorter excution time. Prior systems evaluate on discrete GPUs, while extra work on CPU and DRAM are basically considered free.

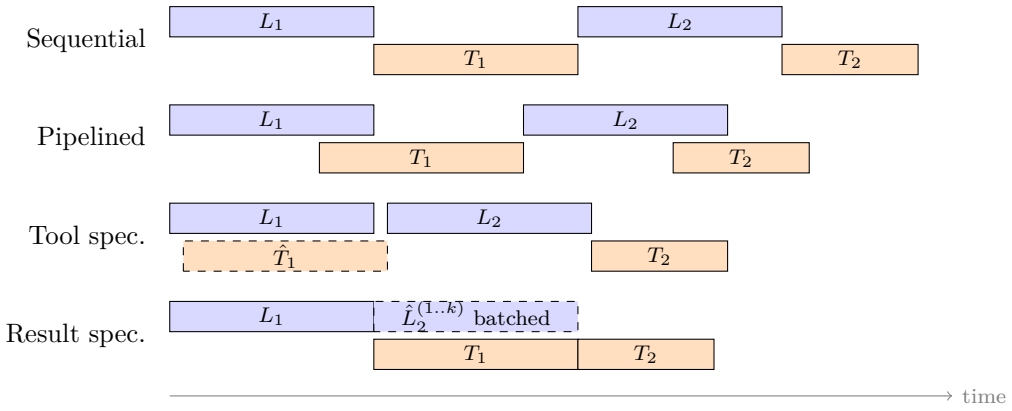


Figure 1: One agent under four schedules. The upper lane (purple) is GPU decode and the lower lane (orange) is CPU tool execution.

How much one agent can gain. At step i, overlap can hide at most $\min(L_i, T_i)$, where L_i is the LLM time and T_i the tool time, so the possible headroom of a request and the best speedup are

$$H \;=\; \frac{\sum_i \min(L_i, T_i)}{\sum_i (L_i + T_i)}, \qquad \text{speedup} \;\le\; \frac{1}{1-H}.$$

We computed H over all 82k TraceLab requests, with L_i from recorded cloud times or local times measured on the M5 Pro(Table 1). We found that tool time concentrates in a few long calls and 40% of requests call no tool at all. With one-step overlap, a single agent gains at most 1.05–1.07×, and up to 1.20× if speculation could look arbitrarily far ahead. Overlap inside only one agent therefore has a clear ceiling, while independent agents running at once have no such dependency limit.

LLM time	Context	Tool share	H (one step)	Bound	H (unlimited)	Bound
Cloud (recorded)	–	32%	6.5%	1.07×	16.8%	1.20×
Qwen3-8B	16k	23%	5.9%	1.06×	15.0%	1.18×
Qwen3-8B	32k	17%	5.2%	1.06×	12.6%	1.14×
Qwen3-8B	64k	12%	4.4%	1.05×	9.7%	1.11×

Table 1: Single-agent headroom over TraceLab. Speedup bound is $1/(1-H)$. "Unlimited" assumes perfect speculation with no lookahead limit.

Many agents on one machine. When several agents run at once, their sessions have no data dependencies on each other, so their work can run in parallel (Figure 2). The GPU decodes every agent that is currently generating as one batch. When fewer than N agents are generating, the freed up batch slots can also run speculative work for agents that are waiting on a tool.

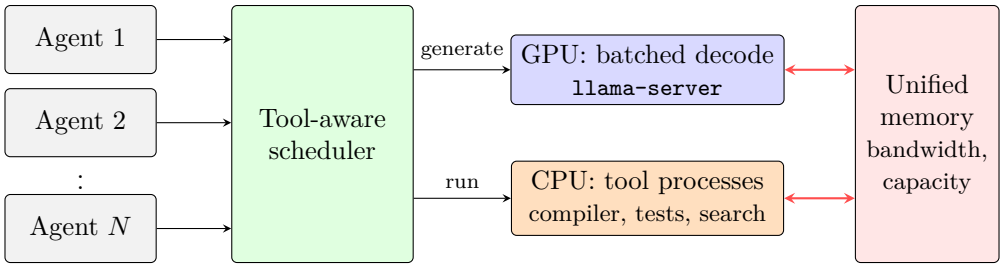


Figure 2: System overview. N independent agents share one machine: the GPU decodes every generating agent in one batch and the CPU runs their tools. Both draw on the same memory, so the scheduler decides when each tool call runs.

Related systems. EdgeAgent [2] schedules several agents on unified memory but models tool calls as pauses, so it does not measure how generation affects the tools. SERENO [3] studies bandwidth contention between LLM inference and foreground apps on phones. We run real tools and batched generation locally on the unified memory.

The Challenge

What we hope to learn. When does parallelizing the agent loop pay off on a machine where the parallel work and the critical path share memory, and what does a scheduler need to know about each tool to make that decision?

Dependencies. The workload is a chain by construction. The LLM must see the tool output before it can decide the next call. The only parallelism available is speculative (we guess the next call) or partial (we start the call before the LLM has finished talking), and both have a cost when wrong. The dependency structure is also data-dependent and only revealed at run time, so static scheduling is impossible. The runtime has to decide whether to speculate and how much. A wrong speculation must leave no trace, but compilers, test runners, and `pandas` scripts write files. Speculative tools therefore run in an isolated copy of the workspace (an overlay or copy-on-write snapshot) whose changes are committed only on a correct prediction, or else are restricted to tools we can show are read-only. Isolation has its own setup cost, which must stay below the latency it hides. A speculative branch is valid only if the tool output it assumed is exactly the text the harness would have appended, since the LLM conditions on tokens, not on outcome classes. Raw outputs contain timings and paths that are unpredictable, so the harness must canonicalize common outcomes (e.g., a fixed message for a clean build or a full test pass) to make them predictable at all. Choosing which outcomes to branch on, and how many (k), trades hit rate against KV capacity.

Memory access and communication. Explicit communication is small. Such as HTTP requests to `llama-server`, `fork`/`exec` and pipes for tools, a per-token CPU–GPU synchronization inside the engine, and cancel/commit messages for speculation. The potentially dominant cost is implicit: the contention for bandwidth and capacity in the shared memory system. On a unified-memory system, the CPU and GPU share one memory pool. Decoding is bandwidth-bound, since every step streams the weights and every batched agent's KV cache through that memory, and many tools also consume significant bandwidth or cache. On a discrete GPU the same tool traffic goes to the host's own DRAM and never touches the GPU's memory. A running tool therefore slows decoding for the whole batch, and decoding slows every running tool, though not symmetrically. We did

preliminary experiments and confirmed this asymmetry. A basic mitigation is to stagger decode away from tools, but that leaves the GPU idle and costs throughput. Different tools have different footprints: `ripgrep` is a streaming read, a compiler is latency-bound with many small allocations, `pandas` is a large working set with random access, `pytest` is a mix. A good policy has to know, per tool, how much it hurts decoding and how much decoding hurts it, in the style of Bubble-Up [4]. We must measure this ourselves; no prior work reports it for agent tools on unified memory.

Agents also interfere with each other. Batching N agents into one decode stream shares the weight reads, but each agent adds its own KV cache to every step. With a long agent context, one agent's KV cache can be as large as the quantized weights, so with only a few agents every decode step reads more KV cache than weights, and each added agent slows the step for all the others. Prefill adds a second kind of interference. When a tool returns, its output (for example, a long test or build log) is appended to that agent's context and must be prefilled before the agent can decode again. `llama-server` puts this prefill in the same batch as the other agents' decode steps, so one agent's long tool output slows everyone's generation. If an agent's KV cache is evicted, for example because there are more agents than server slots, its whole context has to be prefilled again.

Divergence. Speculative branches behave like divergent lanes: the k branches in a batch generate different lengths, so the batch runs until its longest branch finishes or is cancelled. Mispredicted branches are wasted lanes, and a mispredicted tool speculation is wasted CPU work that still occupies the memory system. Tool calls are heavy-tailed: the few long calls dominate total time, and those are the ones most likely to interfere. A policy tuned to the average call is tuned to millisecond calls that contribute almost nothing to total time.

Capacity. On the 24 GB laptop the model weights, the KV cache for a long agent context, and the tool's working set all share one pool. Speculating a `pandas` job that allocates several gigabytes can evict KV cache or force the engine to swap, which slows the run and also corrupts the measurement.

Overhead and measurement. Most tool calls take milliseconds, so the runtime's own overhead (Python `asyncio`, process spawn, HTTP round trips) must stay well below that or it eats the gain. macOS gives no `taskset`/`cgroups`, so core pinning is only available on the GB10. Attributing a slowdown to bandwidth rather than to core contention requires controlled tool variants (sleep-only, CPU-only loop, real tool) and a reference with no contention, which we build from solo runs of the LLM and of each tool on the same machine.

Resources

Hardware.

- NVIDIA GB10 / DGX Spark (128 GB unified memory, 273 GB/s bandwidth, 10P+10E Arm cores, Blackwell GPU, Linux/CUDA).
- Apple M5 Pro laptop (24 GB unified memory, 5P+10E CPU cores, 16-core GPU, Metal).

Software. `llama.cpp` / `llama-server` for inference on both CUDA and Metal. We will write the runtime, simulator, and scheduler in Python and measure their overhead. Tools are unmodified programs such as `ripgrep`, `clang`/`gcc`, `pytest`, or `pandas`. For profiling we use tools such as Nsight Systems and `perf` on Linux, and `time` and Instruments on macOS.

Traces. TraceLab [1] (public) and our own Claude Code session logs (177 sessions, 12.7k tool results). The nebius SWE-agent-trajectories dataset has full commands and observations but no timing.

Starting point. We start from scratch for the runtime, simulator, and scheduler, and reuse `llama.cpp`.

Literature. PASTE [5], Conveyor [6], LLMCompiler [7], SERENO [3], EdgeAgent [2], Bubble-Up [4], TraceLab [1], value prediction [8].

Goals and Deliverables

Baselines.

Solo One agent on the machine: decode and tool call run in order, while independent calls of one step run concurrently. This serves as reference for scaling efficiency.

Naive N agents share one `llama-server` with continuous batching. Every tool runs when issued. This is the baseline we must beat.

Stagger A naive mitigation of tool slowdown by adding decoding pauses whenever a tool runs.

Ideal We measure each decode step and tool alone, then add up these isolated times along each of the N agents' traces to compute the best achievable time, as if there were no contention between agents and no cost of memory sharing.

Metrics.

- Throughput (agent requests completed per unit time) and per-request active time (time from user prompt to final response), as functions of N.
- Scaling efficiency: throughput(N) / ($N\times$ throughput(1)).
- Interference: slowdown of each tool call and of decoding relative to isolated runs.
- Recovered gap: (Naive − Our time) / (Naive − Ideal), for throughput and active time.

Plan to achieve.

1. **Workload model** from traces: per-agent tool share, tool duration by category, token counts per step, and the single-agent headroom H. Done with M5 Pro rates (Table 1); we will repeat the local rates on the GB10.
2. **Tool data cards**: for each tool, average cores used, peak RSS, bandwidth demand, and Bubble-Up-style sensitivity and aggressiveness against decoding at batch sizes 1 to N, on GB10 and M5 Pro, with three variants per tool (sleep-only, CPU-only loop, real). One more card for the LLM: decode slowdown versus the number and type of concurrent tools.
3. **Multi-agent replay runtime**: N sessions against one `llama-server`, fixed output lengths (`n_predict` + `ignore_eos`), a shared system-prompt prefix, tool substitution by category and duration, and its own overhead measured.
4. **Interference-free simulator**: event-driven replay of the same N agents using solo timings, validated against measurement at $N = 1$, where there is no interference between agents.
5. **Scaling study**: sweep N on both platforms for Naive and Stagger and compare against Ideal; find where scaling stops on each machine and whether capacity or bandwidth stops it.
6. **Tool-aware policy**: use the data cards and the current load to decide, per tool call, whether to admit it, delay it, or slow generation to protect it, together with a cap on N and core placement on the GB10; compare against Naive, Stagger, and Ideal.

Targets.

- Recover at least 50% of the Naive–Ideal gap in p95 active time at $N = 8$ on the GB10, with throughput within 10% of Naive. A few long, bandwidth-heavy tools should cause most of the interference, so acting only on them keeps most of the overlap.
- Match Stagger's p95 active time with at least 20% higher throughput, because pausing decode only for contending tools leaves the GPU idle far less often.
- Simulator error within 10% at $N = 1$, where nothing overlaps and only runtime overhead remains.

Hope to achieve.

1. **Speculation in the batch slack**. When fewer than N agents are generating, the free batch slots can run speculative work for agents that are waiting on a tool, such as result speculation: decode k branches, each conditioned on a likely tool outcome, and keep the branch whose assumed output matches. A branch can be committed only if its assumed output is exactly the text the harness would append, so common outcomes must be canonicalized, and speculative tools must run in an isolated copy of the workspace. Speculative branches are nearly free when the batch has room and take bandwidth from real work when it does not, so the accuracy at which speculation breaks even should depend on load. The headroom study caps the single-agent gain at a few percent, so we treat this as a question about load, not about latency.
2. **A third unified-memory platform** (Mac Studio) to test whether the policy transfers.
3. **An online version of the policy** that senses contention from decode latency, as in SERENO.

If progress is slower.

1. Workload model, tool data cards, the replay runtime, the simulator, and the N sweep for Naive and Stagger on both platforms. The deliverable is then the characterization: where scaling stops on each machine and why.
2. If replaying real tools proves unreliable, we replace them with synthetic tools (sleep, CPU loop, bandwidth kernel) calibrated to the data cards.

Poster. Four plots: throughput and p95 active time versus N on both platforms, with Naive, Stagger, and our policy measured against Ideal; an interference matrix (tool category $\times$ batch size); a timeline of N replayed agents under Naive and under the policy; and the single-agent headroom that motivates the multi-agent setting. We will also show a live replay on the laptop.

Platform Choice

Our question is about machines whose CPU and GPU share memory, so we use two unified-memory machines that cover both vendors and operating systems that ship them. They differ by about $5\times$ in memory capacity, so we expect the M5 Pro laptop to run out of KV-cache space first and the GB10 to saturate bandwidth first. We use `llama.cpp` because it runs on both CUDA and Metal with the same quantized weights, which removes the engine as a variable.

Development order. We develop and debug on the M5 Pro laptop and use the GB10 mainly as the measurement machine. We check correctness under every policy: a replayed agent should produce the same tool outputs, the same token counts per step, and the same final transcript whether it runs alone or with $N - 1$ others. Only timing may differ between runs and platforms.

Schedule

Dates	Goal
Nov 9–12	Build the workload model from traces and measure single-agent headroom (done on laptop already). Measure prefill and decode rates versus context length and batch size.
Nov 13–15	Profile how each tool uses cores, memory, and bandwidth, alone and concurrent with batched generation.
Nov 16–19	Build the multi-agent runtime and check that every agent's transcript is identical alone and in a group.
Nov 20–22	Build the ideal baseline and validate it at $N = 1$. Write the milestone report.
Nov 23	**Milestone report due.**
Nov 23–26	Profile the tools on the GB10. Sweep N with naive concurrency on both machines.
Nov 27–29	Add the stagger baseline. Measure the gap to the simulator on each machine and profile it with Nsight Systems, `perf`, etc.
Nov 30–Dec 2	Build and refine the policy.
Dec 3–5	Add the limit on how many agents could run at once and to place core placement on the GB10. Compare the policy against Naive, Stagger, and Ideal.
Dec 6–8	Try speculation in the batch slack. Strengthen experiments and statistic confidence.
Dec 9–10	Write the final report.
Dec 10	**Final report due.**
Dec 11	**Poster session.**

References

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